

- Luminosity of stars
- HR diagram → periodic table of Astronomy



Luminosity of star or absolute magnitude = energy output/s

Luminosity is power (Watt = joules/second = number of photons/s), energy produced per unit time.

There is a relationship between absolute magnitude and luminosity
They both depend on energy output

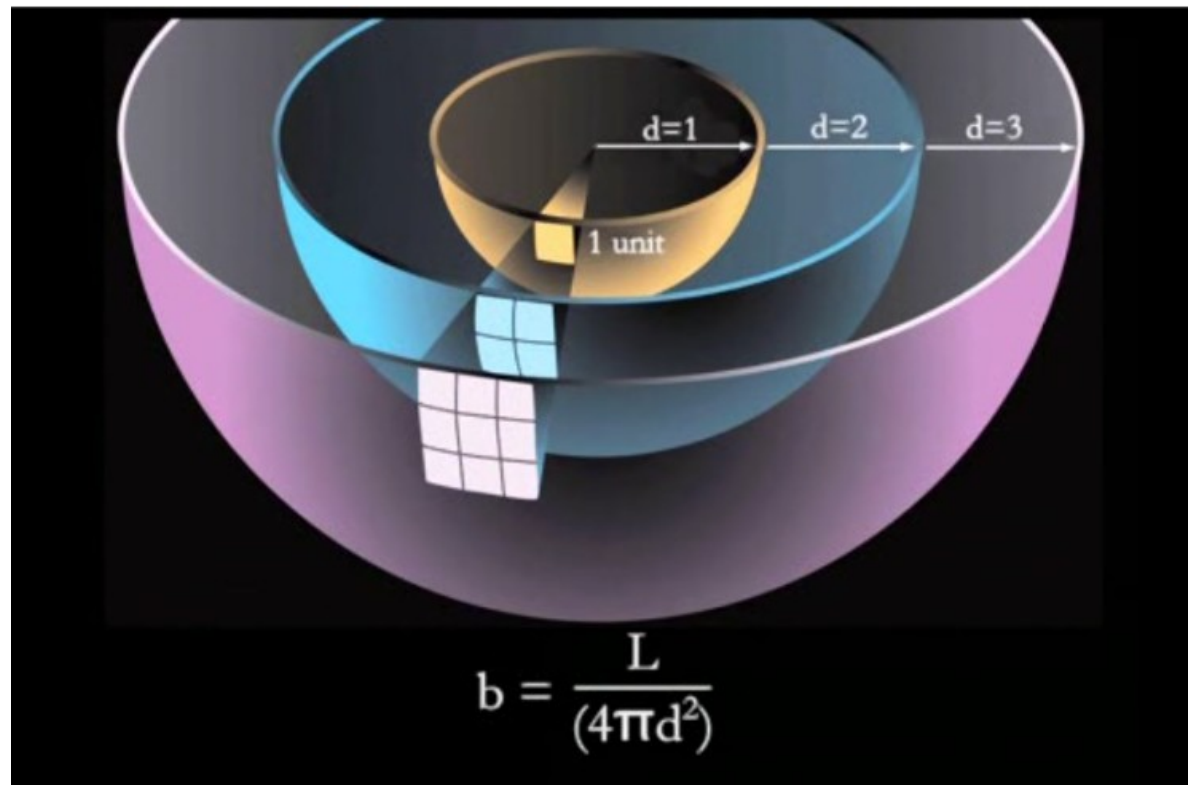
Apparent magnitude or brightness = energy output /s per unit area

Brightness is power per unit area (Watt/m²=joules/s/m²).

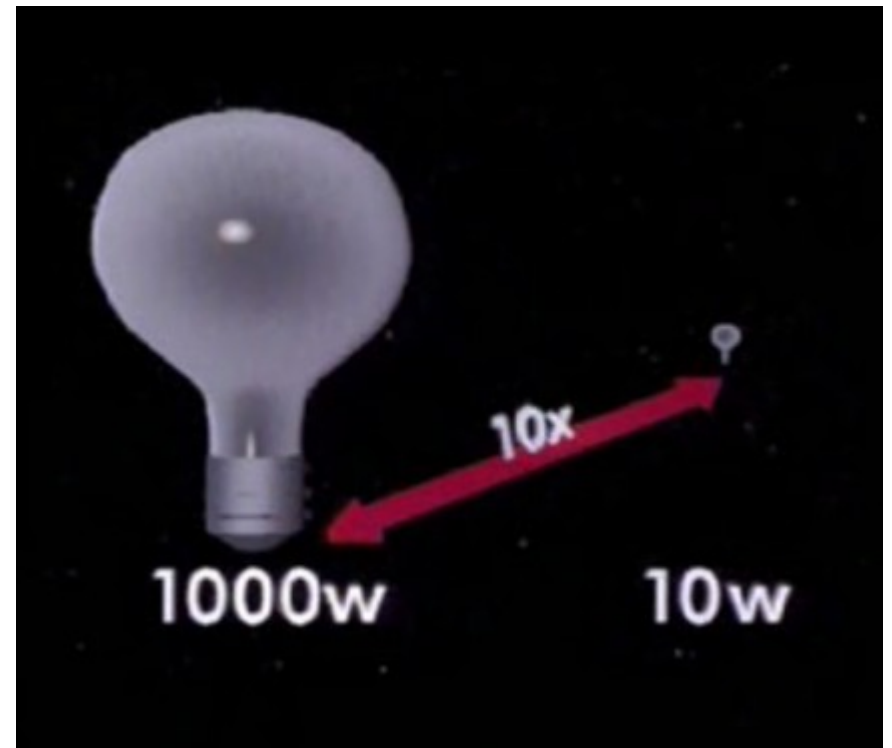
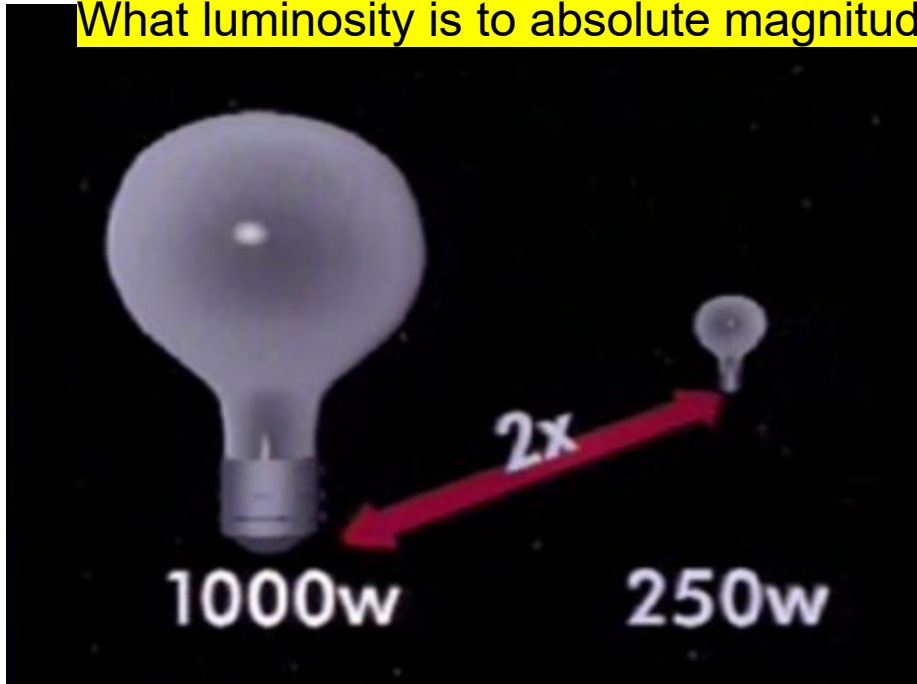
It depends on the distance between stars and observer.

There is a relationship between brightness and apparent magnitude.

They both depend on
Observed brightness.



Today we use more Luminosity versus brightness
And not absolute magnitude versus apparent magnitude
Brightness is to apparent magnitude
What luminosity is to absolute magnitude



Luminosity (power, “oomph”)

$$b = \frac{L}{4\pi d^2}$$

b = apparent brightness

L = luminosity

d = distance

4×10^{26} W light bulb = Sun

One “Solar Luminosity”

$$L_{\odot} \approx 4 \times 10^{33} \text{ ergs/sec}$$



When Sun light get to the Earth
→ 1000 W / m²



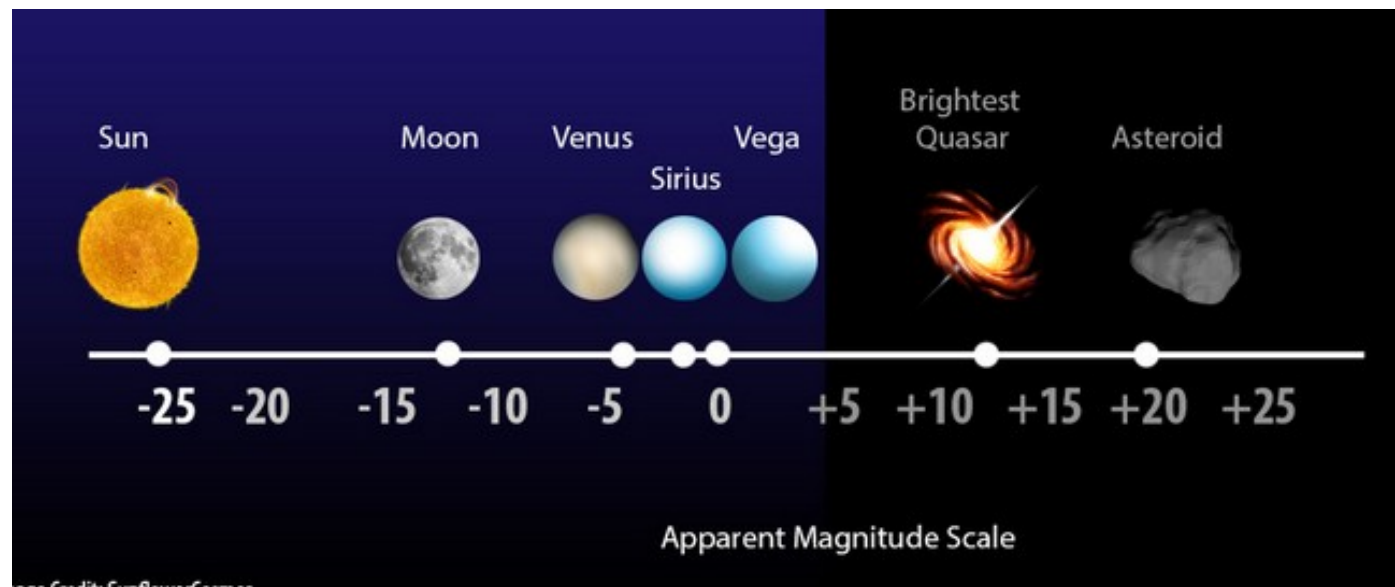
Hipparchus, Greek
Astronomer cataloged
150 stars according to
Their apparent brightness.

Brighter star → magnitude 1

Fainter star → magnitude 6

So brighter → smaller magnitude
(number photons per second and per unit area)

Object	Apparent Magnitude
Typical brightest stars	1
Faintest stars seen with unaided eye	6
Pluto	14
Faintest stars visible with great telescopes	30



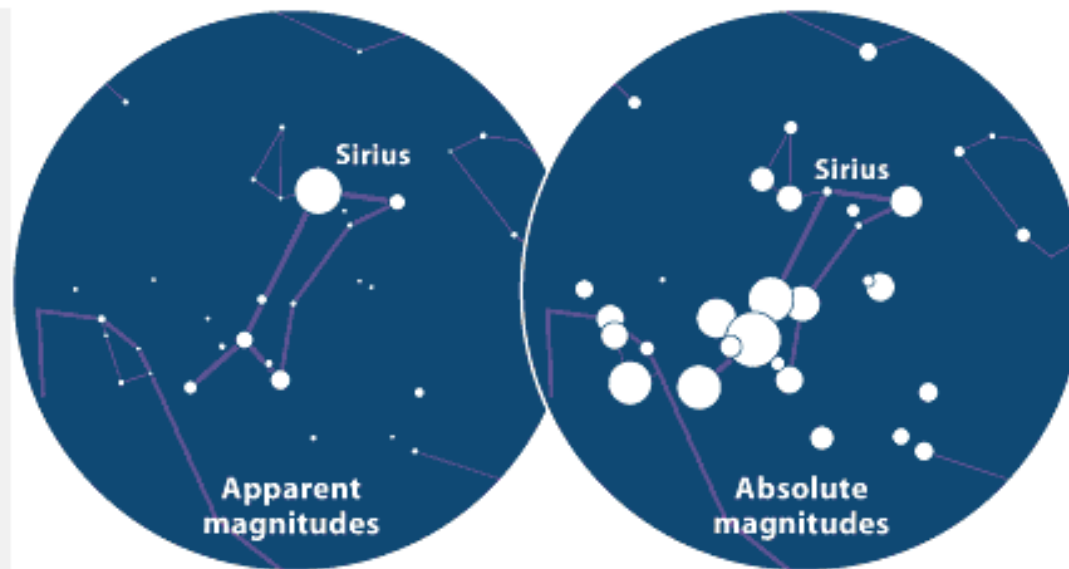


0.12 Rigel, Betelgeuse 0.5
Belt stars 2



Sirius brighter star = -1.5

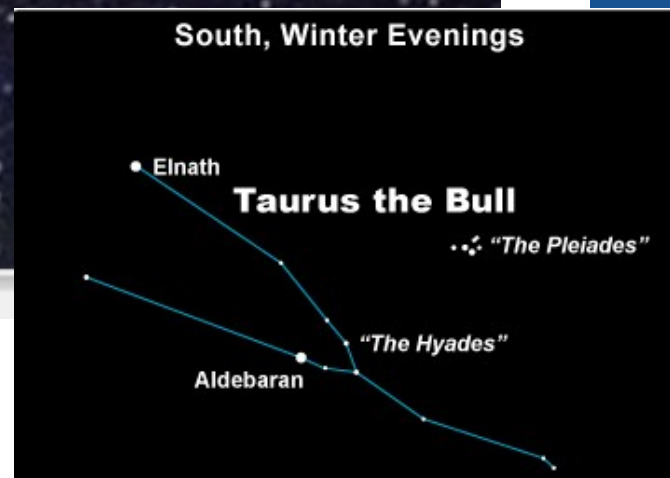
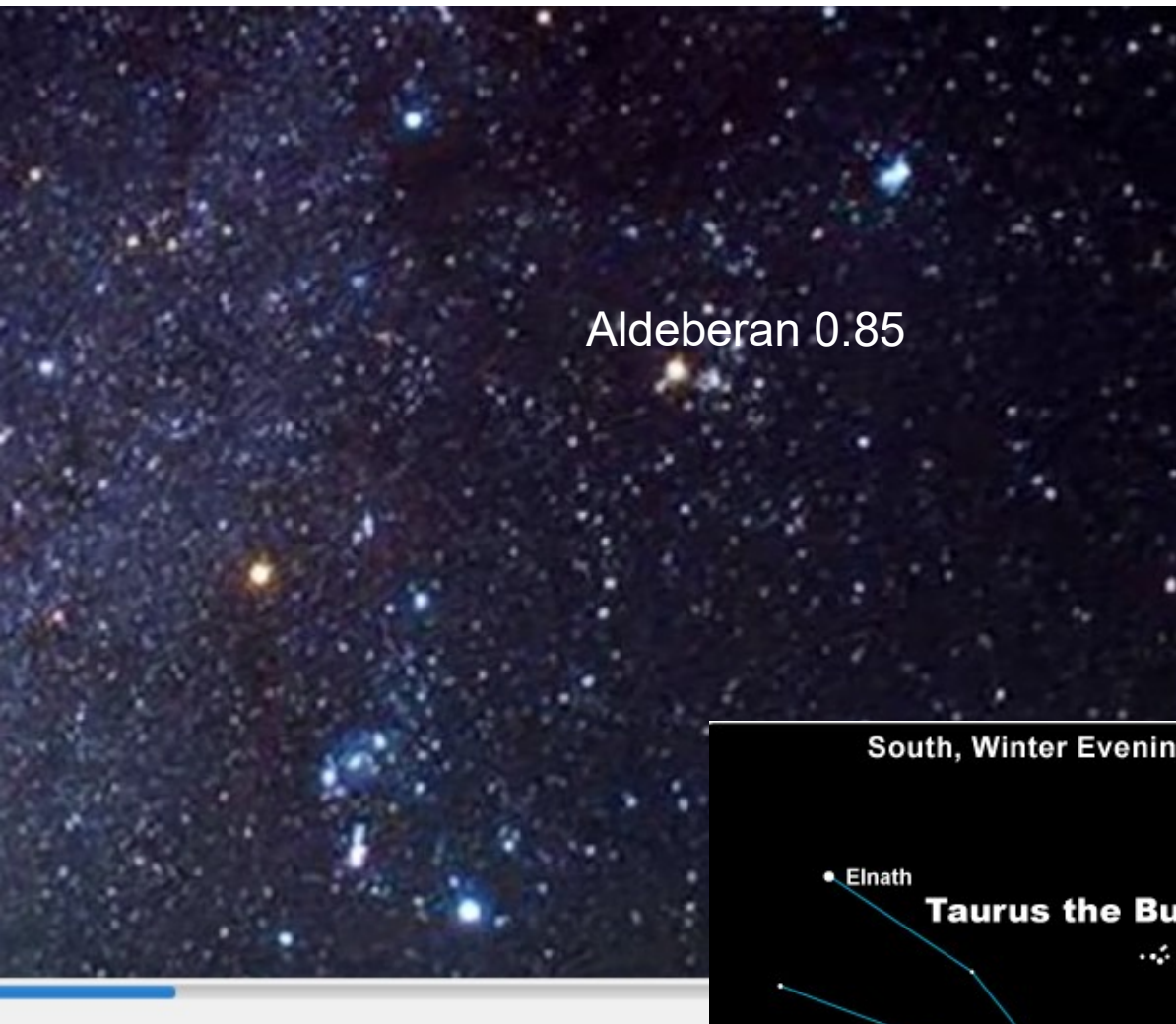
Sirius, also known as the Dog Star or Sirius A, is the brightest star in Earth's night sky. The name means "glowing" in Greek — a fitting description, as only a few planets, the full moon and the International Space Station outshine this star. The bright component of the binary is a blue-white star 25.4 times as luminous as the Sun

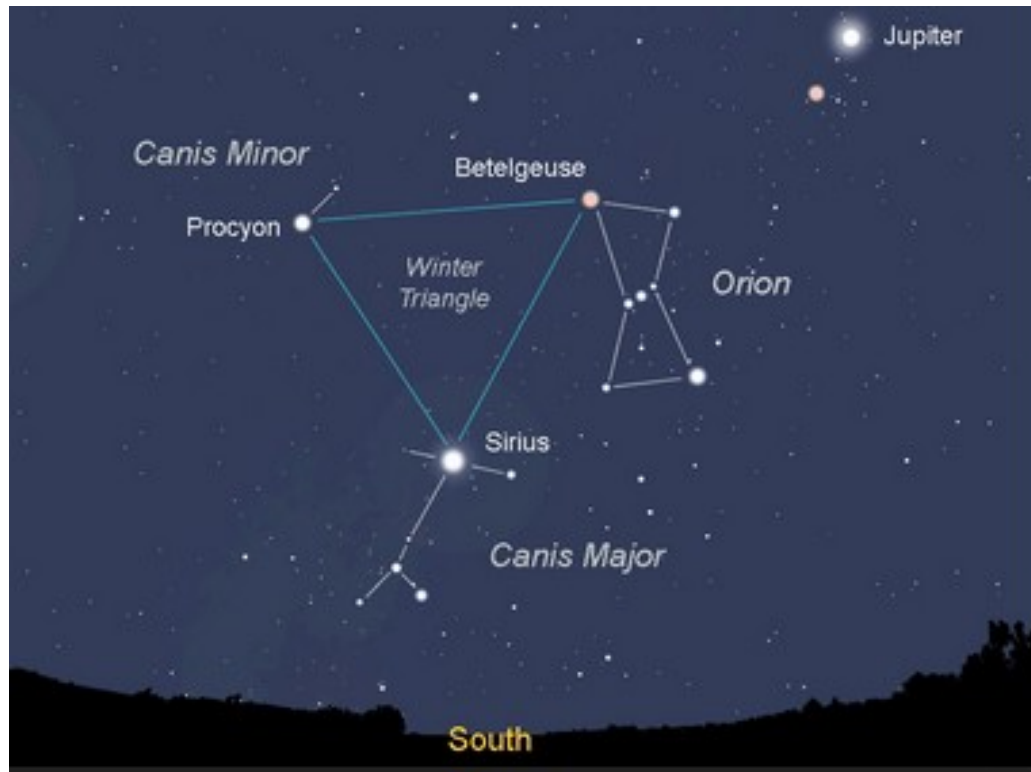


On the left-hand map of Canis Major, dot sizes indicate stars' *apparent magnitudes*; the dots match the brightnesses of the stars as we see them. The right-hand version indicates the same stars' *absolute magnitudes* — how bright they would appear if they were all placed at the same distance (32.6 light-years) from Earth. Absolute magnitude is a measure of true stellar luminosity.

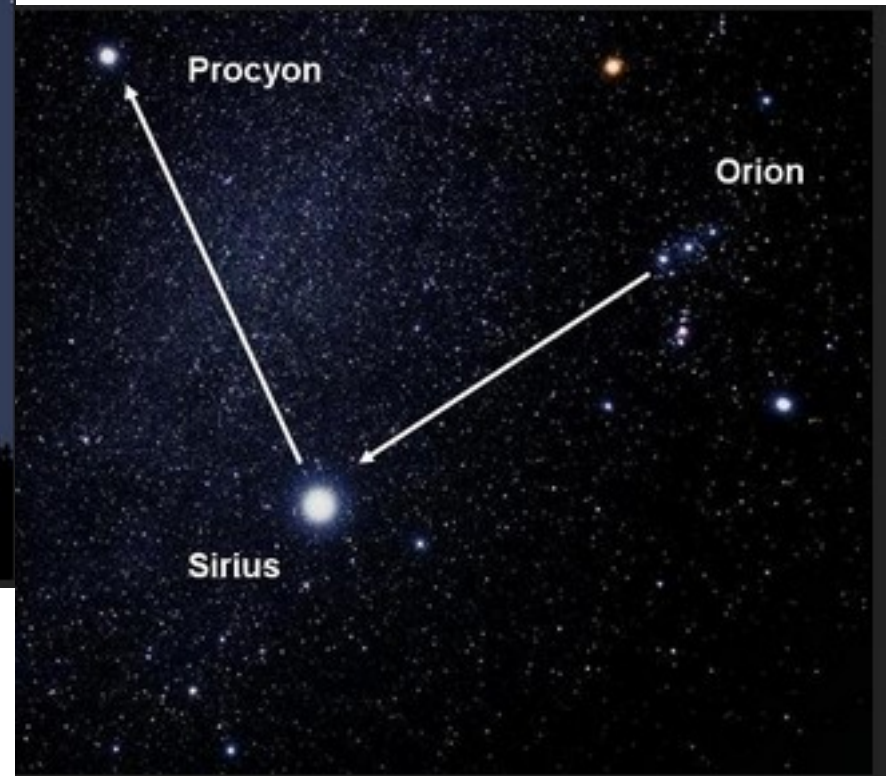
Sky & Telescope

Aldebaran is the brightest star in the constellation Taurus and is known as the "Eye of Taurus." The name "Aldebaran" is Arabic, which means "The Follower" because it appears to follow the Hyades star cluster that forms the head of the bull. red giant star = eye of the taurus



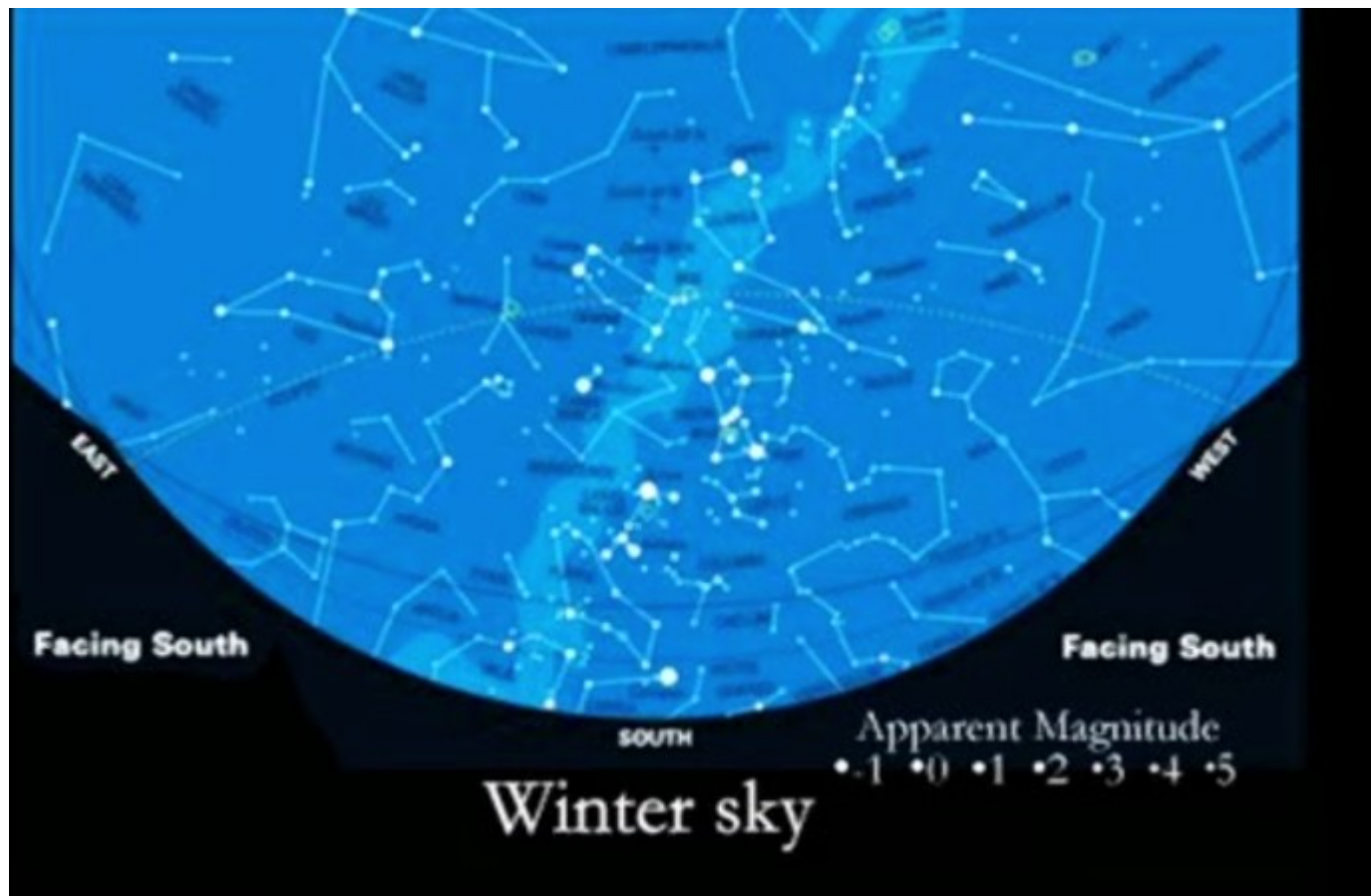


Procyon 0.38

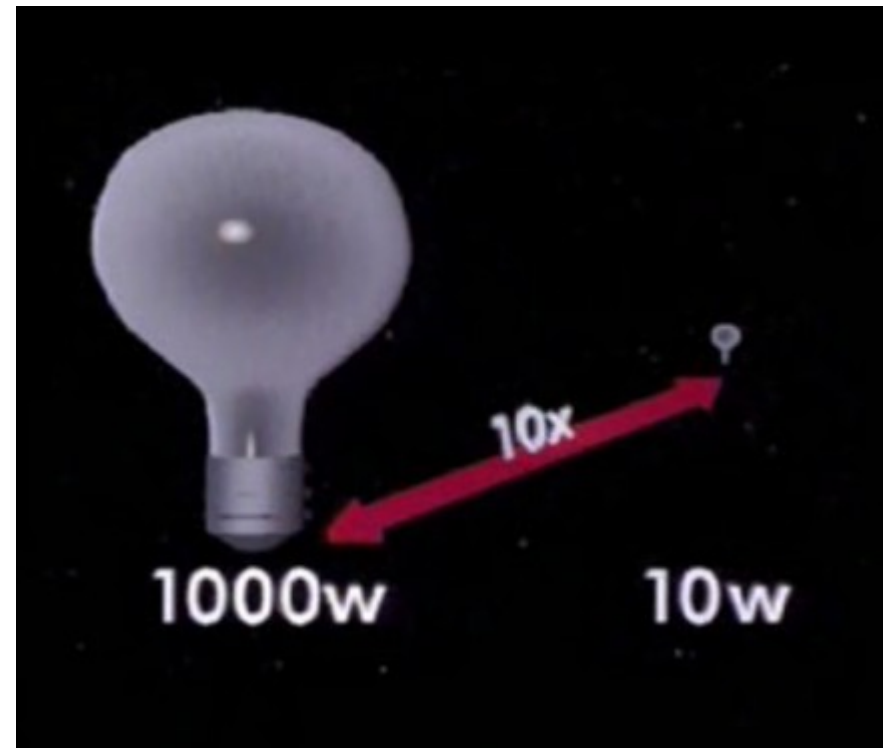
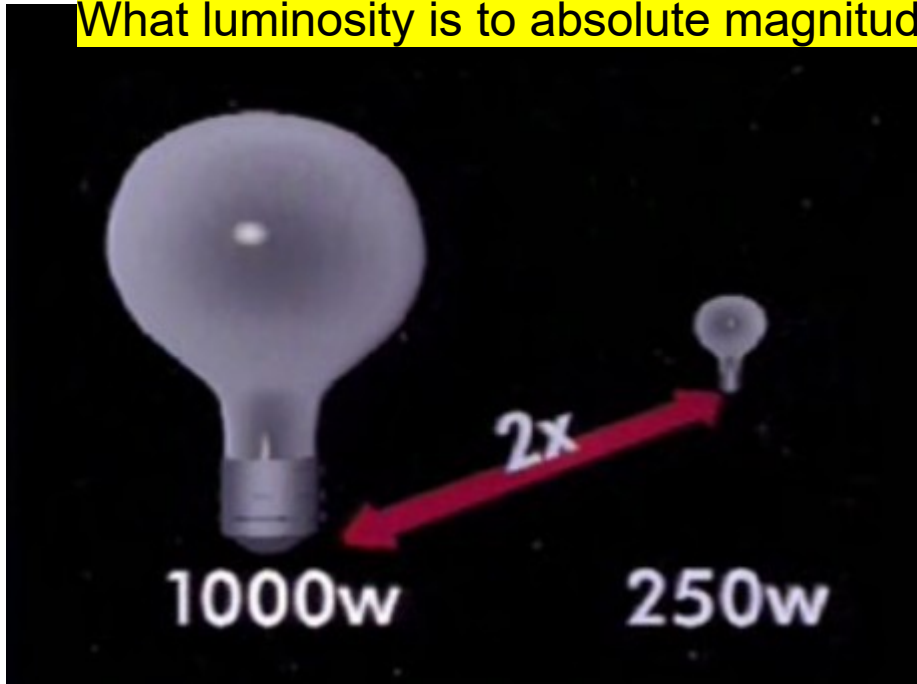


The system is made up of Procyon A (a main sequence star) and Procyon B (a much smaller white dwarf.). It is a F star. (white)

Better image. Star loator



Today we use more Luminosity versus brightness
And not absolute magnitude versus apparent magnitude
Brightness is to apparent magnitude
What luminosity is to absolute magnitude



Luminosity (power, “oomph”)

$$b = \frac{L}{4\pi d^2}$$

b = apparent brightness

L = luminosity

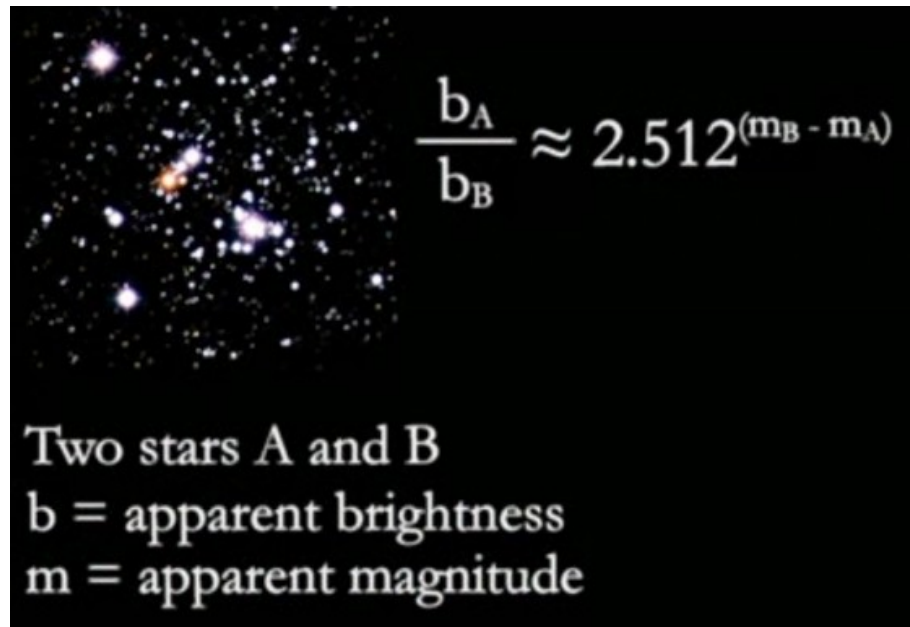
d = distance

Relationship brightness and apparent magnitude (based on our eyes perception)

its a log scale. When magnitude increases

By 1, brightness is multiplied by 2.5 (its actually $(100)^{1/5}$ The fifth root)

Brightness is power per unit area



From $m_1=6$ to $m_2=1$

Brightness is multiplied by 100

And $m_1 = m_2+5$ (magnitude decreases by 5)

(so 100 more photons hitting your eyes)

So it is a log scale.

$$m_2 - m_1 = 5/2 \log(b_1/b_2) = -5/2 \log(b_2/b_1)$$

If $b_2 = 100$ $b_1=1$ then $b_2/b_1= 100$

Then $\log(100) = \log(10^2) = 2$

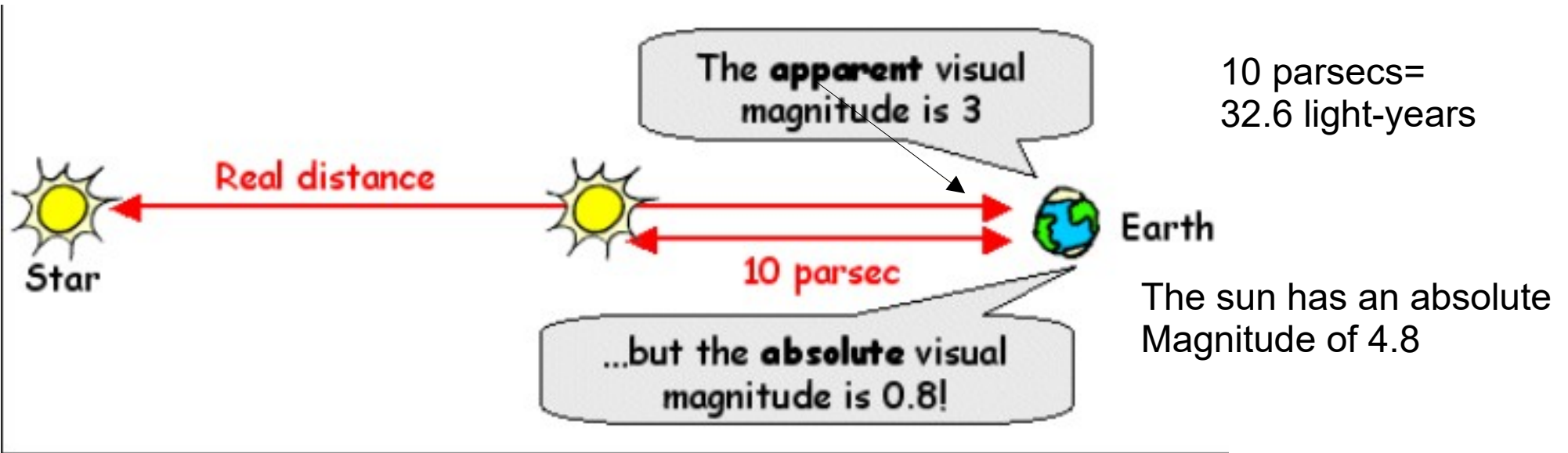
$-5/2 \log(b_2/b_1) = -5$

$m_2 - m_1 = -5$

$m_1 - m_2 = 5$

What is the absolute magnitude M?

Absolute magnitude is to magnitude what luminosity is to brightness



$$m - M = 5 \log \left(\frac{d}{10} \right)$$

d = distance in parsecs

m = apparent magnitude

M = absolute magnitude

This equation is derived from the inverse square law. Brightness is proportional to luminosity divided by distance squared.

$m_2 - m_1 = 5/2 \log(b_1/b_2) = -5/2 \log(b_2/b_1)$

$m_2 = m \rightarrow$ apparent brightness b_2 of a star located at distance $d_2 = d$ pcs (luminosity L).

$m_1 = M \rightarrow$ absolute brightness b_1 of the same star located at a distance $d_1 = 10$

$b_2/b_1 = L_2/L_1 \times (d_1/d_2)^2 = (d_1/d_2)^2 = (10/d)^2$ as $L_1 = L_2 = L$.

So $m - M = -5/2 \log((10/d)^2) = -5 \log(10/d)$

Of $m - M = 5 \log(d/10)$. d is in pcs.

Stefan-Boltzmann Law

The **Stefan-Boltzmann Law** tells you how much energy per unit area is emitted from a black body.

$$\varepsilon = \sigma T^4 \quad (\sigma = \text{Greek letter "sigma" = constant})$$



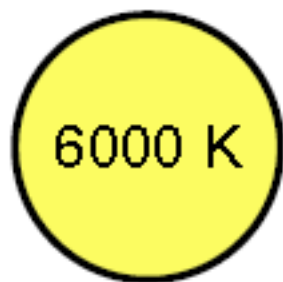
Area in cm^2

Energy per area per sec

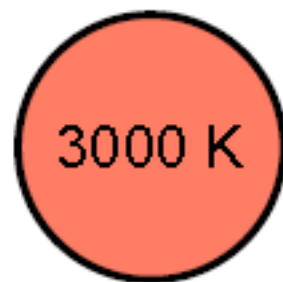
Because of the T^4 power, hotter object emit MUCH more energy per area than cold objects.

Big change in brightness for small change in temperature !

Example: always work with ratios !



6000 K



3000 K

$$\left(\frac{6000}{3000}\right)^4 = 2^4 = 16$$

If a star is 2x as hot, it emits 16 times more power! (if the two stars have the same surface area)

Luminosity (or Power)

This is the amount of energy emitted per second by the **entire** object.

$$L = (\text{Surface Area}) \times \left(\frac{\text{energy emitted}}{\text{cm}^2 \text{ sec}} \right)$$

$$L = (\text{Surface Area}) \times \sigma T^4$$

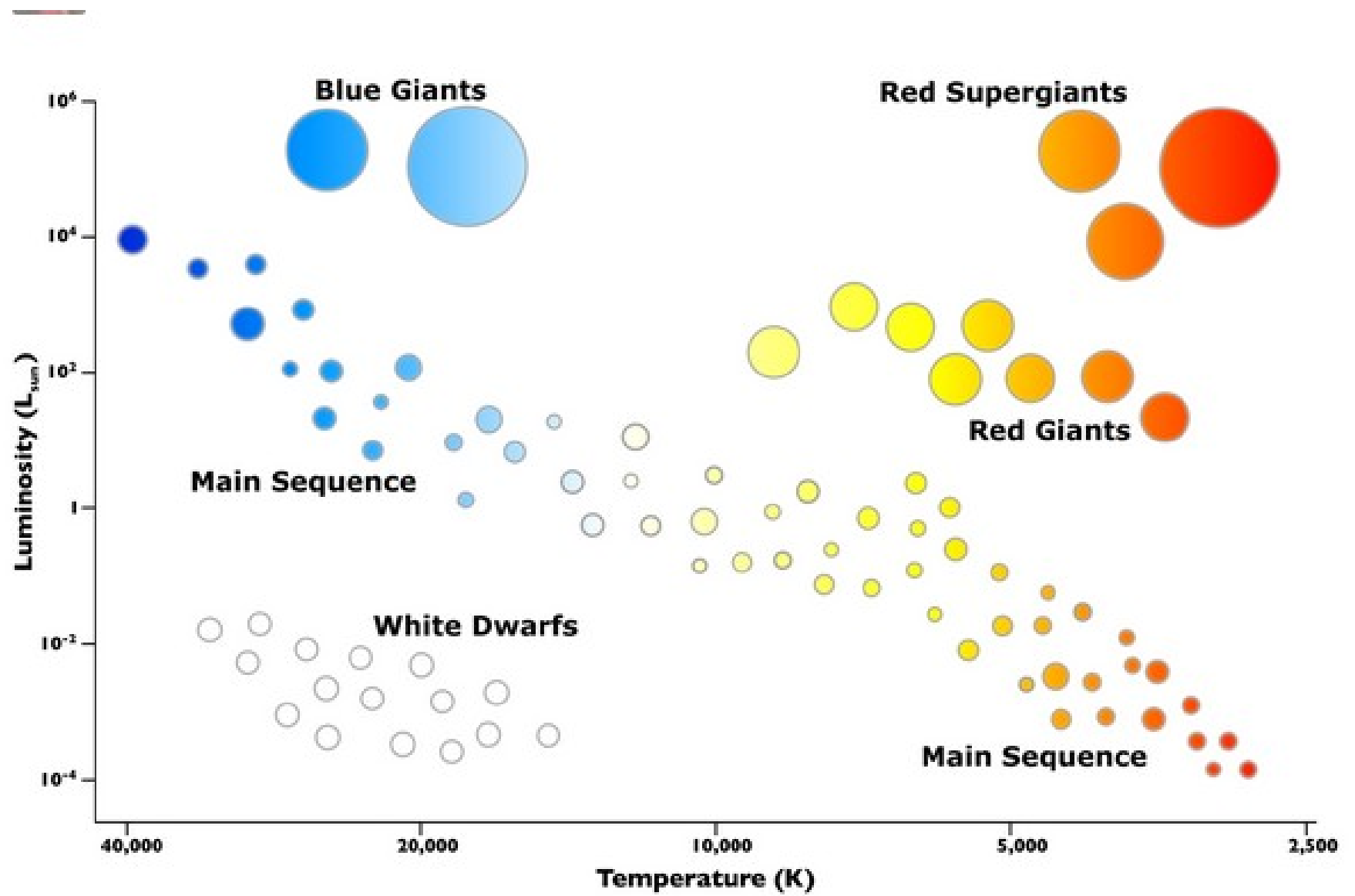
For a sphere of radius R , Surface Area = $4\pi R^2$, so that

$$L = 4\pi R^2 \sigma T^4$$

If we know both L and T , then we can derive the radius, R !

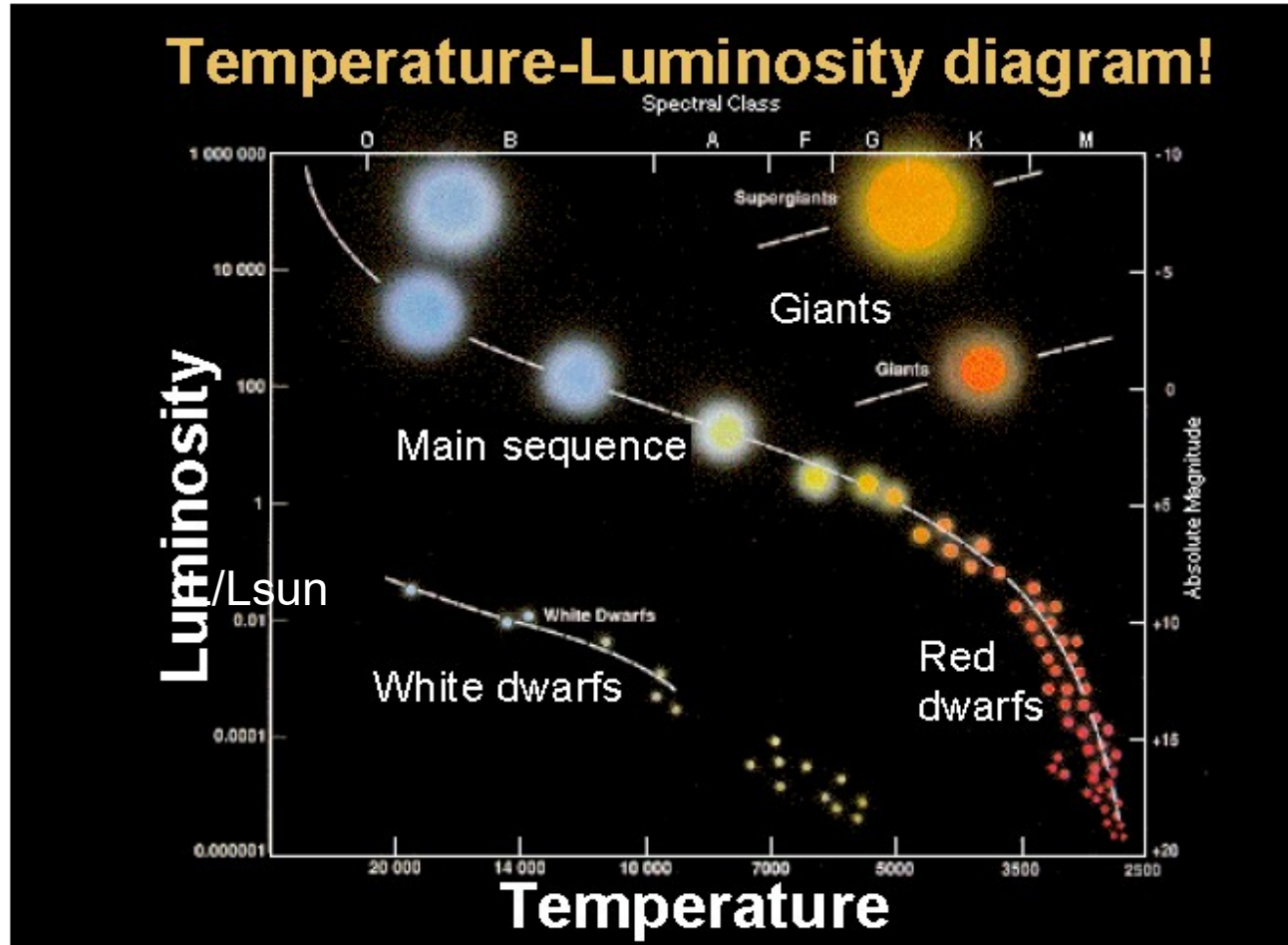
SO IF you increase the size of the object, you increase its LUMINOSITY (brightness). So if a STAR is large enough, it can OUTPOWER a hotter but SMALLER star

Stop → short movie and picture sizes



Hertzsprung-Russell diagram (HR)

shows the relationship between Temperature and luminosity.



The Hertzsprung–Russell diagram



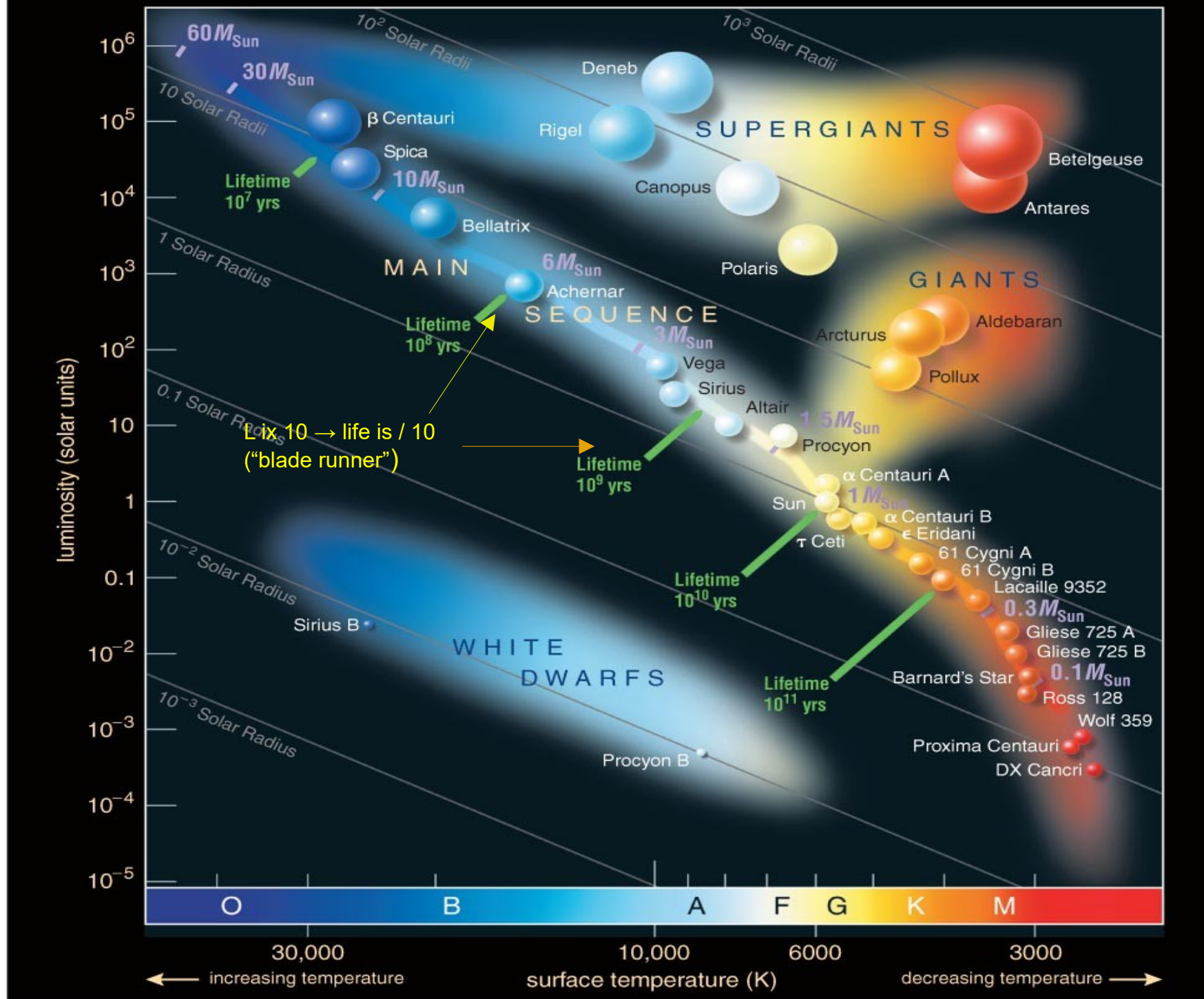
Ejnar Hertzsprung
Henry Norris Russell

Used to find distance of
clusters

Age of clusters

Periodic table of astronomy

Luminosity ↑



HR diagram(Hertzsprung-Russell). Notice the log scale on y-axis

See **movie** about HR diagram

Main feature of the T-L diagram

- **Main sequence:** Stars that are happily undergoing nuclear burning of their fuel (like our Sun). This is where a star spends most of its life.
- **Giants:** Stars that have left the main sequence because their radii expanded. This is a shorter, later phase of a star's life.
- **Red dwarfs:** Small, cooler stars on the main sequence. They have very long lives of 100 billion years or more.
- **White dwarfs:** Blue or white stars with very small radii. These are basically dead stars that aren't burning fuel anymore.

Video HR diagram.

Video Sun evolution

